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SCIENCE

General Science Foundations

Class 6

Author: NCERT Expert Panel

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Chapter 1

Matter in Our Surroundings

Matter is anything that occupies space and has mass. It exists in three primary states — solid, liquid and gas — distinguished by the arrangement and movement of particles. Solids have fixed shape and volume, liquids flow and take the shape of the container, while gases expand to fill available space. Changes of state such as melting, freezing, evaporation and condensation involve energy exchange. Plasma and Bose-Einstein condensates are additional states of matter. Understanding particle nature explains phenomena like diffusion, compression and the behavior of substances under varying temperature and pressure.

Chapter 2

Motion and Force

Motion is a change in the position of an object with respect to time. Displacement, velocity and acceleration describe motion quantitatively. Newton's three laws of motion form the foundation of classical mechanics: the law of inertia, the relation between force and acceleration ($F = ma$), and the action-reaction principle. Forces can change the state of motion or the shape of objects. Friction, though often a hindrance, is essential for walking, braking and gripping. Understanding motion and force enables analysis of everything from falling apples to orbiting satellites.

Chapter 3

Diversity in Living Organisms

Living organisms display enormous diversity in form, habitat and function. Classification organizes this diversity into groups based on shared characteristics. The five-kingdom classification by Whittaker divides organisms into Monera, Protista, Fungi, Plantae and Animalia. Within animals, further division into invertebrates and vertebrates reflects evolutionary complexity. Biodiversity ensures ecosystem stability and provides resources for food, medicine and materials. Conservation of biodiversity is vital for sustaining life on earth.